When Field Exclusion Fails: Lineage-Certified Admission for Integrated-Data Benchmarks

Extended Audit Details

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The paper is self-contained; this document lists extended tables and checks of the artifact and is not needed to follow or assess the paper. It gives the certificate-path rule, the exact verification of the Path B results and the mutation checks (Section A), the registry maps (Section B), further results (Section C), a description of the released material (Section D) and the support sensitivity of Path B (Section E). Roman section numbers and unprefixed table and figure numbers refer to the paper, whose notation we use; DF and FP denote DeepFashion and Fashionpedia.

A. Details for Sections III and IV

A.1 Certificate-Path Selection (Remark 2)

Algorithm S1 is the rule of Remark 2; it reads only the exposed blocks and the edge declarations of L^{rel}. In L^{rel}, distances to a carrier differ across blocks, so no audited view has two shortest paths; line 2 is a tie-break for registries that do.

Algorithm S1 Schema-level certificate-path selection

Require: exposed blocks $E(V)$; carrier block c of q; registered edges Π (`core::→facet::`), T (`facet::→type::`), W (`facet::→family::`)
Ensure: path, basis b, required maps $R(V, q)$
1: $P \leftarrow$ shortest registered path from a fully exposed block of $E(V)$ to c (length 0 if $c \in E(V)$)
2: on equal length, prefer `facet::` over `core::` as the start
3: **if** P exists **then return** (A, start of P, edges of P)
4: **else if** c is `facet::` or a registered image of it, and every block of $E(V)$ is a registered image of `facet::` or a coordinate projection of one **then**
5: **return** (B, `facet::`, edges from `facet::` to c and to each block of $E(V)$)
6: **else return** (C, none, $\emptyset$)

A.2 Exact Verification of the Path B Results

Table S1 lists all 10 Path B UNSAFE certificates. Take the type view of the family task in FP and its observed value $v = (\text{attribute}, \text{category}, \text{part}) = (0, 2, 0)$. The basis is `facet::` on S_{FP} and $R(V, q) = \{T, W\}$. A state of the fiber places exactly two events on category facets of S_{FP} and none on attribute or part facets. The FP support contains nine category facets (Table S2), so the facet blocks of its states take $\binom{9+1}{2} = 45$ values. Since W sends every category facet to TAX, every state has the carrier (TAX 2, all other families 0); TAX is the strict unique maximum with margin 2, and every rival program has optimum -2.

The two certificates verified by argument belong to the type{attribute, part} view with value $(0, 0)$ in DF and FP: both exposed counts are zero, so every event of a state lies on a category facet, the number of events is unbounded, and TAX is the strict unique maximum at every state with $\sum_k (x_b)_k \geq 1$.

For each of the 659 open single-label fibers, we record the stopping rival and a state at which it ties with or exceeds the observed label, checked in integer arithmetic (marginals, support, counts and, for an unbounded program, a free facet of the rival).

TABLE S1
EXACT VERIFICATION OF THE 10 PATH B UNSAFE CERTIFICATES; EVERY CERTIFICATE PASSES. VALUE: TYPE COUNTS (ATTRIBUTE, CATEGORY, PART) FOR THE TYPE VIEW OF THE FAMILY TASK, (ATTRIBUTE, PART) FOR TYPE{ATTR., PART}, AND (STR, SURF, TAX, ATTRIBUTE, CATEGORY, PART) FOR FAMILY-3+TYPE OF THE FACET TASK. VALUES: BASIS-BLOCK VALUES OF THE FIBER, EACH CHECKED; AN UNBOUNDED FIBER IS CHECKED BY THE ARGUMENT OF SECTION A.2. MARGIN: SMALLEST GAP BETWEEN THE CERTIFIED LABEL AND THE LARGEST RIVAL OVER THE STATES OF THE FIBER.

View	Set	Value	Label	Method	Values	Margin
Family-3+type	FP	(0,0,1,2,1,0)	waistline	enumeration	9	1
Type	DF	(0,1,0)	TAX	enumeration	3	1
Type	FP	(0,0,1)	STR	enumeration	3	1
Type	FP	(0,0,2)	STR	enumeration	6	2
Type	FP	(0,1,0)	TAX	enumeration	9	1
Type	FP	(0,2,0)	TAX	enumeration	45	2
Type	FP	(1,0,2)	STR	enumeration	72	1
Type	FP	(1,2,0)	TAX	enumeration	540	1
Type{attr., part}	DF	(0,0)	TAX	argument	unbounded	1
Type{attr., part}	FP	(0,0)	TAX	argument	unbounded	1

A.3 Construction of the Mutation Checks

The five mutations of Section VI-A test the auditor on changes whose effects are known in advance; each uses a fixed seed. M1 appends the block $\eta = x_{\mathrm{facet}} G$ with $G = I + \Delta$, where Δ is a nonnegative integer strictly upper-triangular matrix supported on the 15 facets shared by both dataset supports, and registers $(\eta \to \texttt{facet::}, G^{-1})$ in a mutation-specific copy of the registry. The matrix G is unimodular, so G^{-1} is integral and Corollary 1, which needs only $C(x) = x_b \Phi$, applies. M2 remaps 12 of the 1,181 core identifiers to the authority-only facet. M3 appends to `inst_sem::`

a column that is constant within each dataset. M4 applies a positive diagonal rescaling, a coordinate permutation and binary presence to the family-3 and type blocks of the family-3+type view. M5 permutes the labels within each dataset; the check compares the label condition on all 20,204,288 view–anchor pairs, with 150,216–272,155 anchors failing it in each of the four direct-carrier view–dataset pairs, of which the 122,975–166,353 that carry a certificate conflict with their permuted label.

B. Registry Maps (Sections III and V)

TABLE S2
REGISTRY MAPS. T (FACET TO TYPE) AND W (FACET TO FAMILY) ON THE 27 AUTHORITY FACETS, WITH MEMBERSHIP IN THE DATASET SUPPORTS $S_i = \Pi(K_i)$. $T(f)$ IS THE TERM TYPE; `TYPE::` MATERIALIZES THE ATTRIBUTE, CATEGORY AND PART TYPES. THE FAMILY `OTHER` LIES OUTSIDE THE LABEL SET. *THE RELEASED FAMILY BLOCK MATERIALIZES COLOR AS ATTR.

Facet f	$T(f)$	$W(f)$	S_{DF}	S_{FP}	Note
attribute type	attr type	other			authority only
fit	attribute	STR	•	•	
garment detail	attribute	STR	•	•	
leather	attribute	SURF	•	•	
length	attribute	STR	•	•	
neckline type	attribute	STR	•	•	
nickname	attribute	TAG	•		
non textile material type	attribute	SURF	•	•	
opening type	attribute	STR	•	•	
silhouette	attribute	STR	•	•	
textile finishing manufacturing techniques	attribute	SURF	•	•	
textile material type	attribute	SURF	•	•	
textile pattern	attribute	SURF	•	•	
waistline	attribute	ATTR	•	•	
arms and hands	category	TAX		•	
head	category	TAX		•	
legs and feet	category	TAX		•	
lowerbody	category	TAX	•	•	
neck	category	TAX		•	
others	category	TAX		•	
upperbody	category	TAX	•	•	
waist	category	TAX		•	
wholebody	category	TAX	•	•	
color	color	SURF*			no active core term
closures	part	STR		•	
decorations	part	STR		•	
garment parts	part	STR		•	

Registry. The `IN_FACET` target of every authority term equals its facet field (2,715 terms). Table S2 records four cases that the paper uses. Color has nonzero counts in DF but is not in S_{DF}, and the materialized family block sends it to ATTR rather than to SURF; this yields the DF defects of the views that require only W (Table V). The nickname facet lies in S_{DF} but not in S_{FP} and is sent to TAG, a label no evaluated target carries; dropping it on the shared projection breaks FP→DF tie fibers. The authority-only facet is sent to `other`, outside the label set; it lies in neither support, so W sends every facet of S_i to a family label, as the argument after Definition 2 requires. The FP support contains all nine category facets, and W sends each of them to TAX, which the Path B certificates of Section A.2 use.

C. Additional Results for Section VI

All views and bounds. Fig. S1 extends Table IV to all 32 audited views. Cosine predictions fitted on all source anchors are identical when the tolerance for zero or tied scores, 10^{-12}, is raised to 10^{-9} or 10^{-8}. All 96 probe scores lie under their F1 bounds (Fig. S2). On the 275,204 DF and 342,182 FP anchors on which W holds, the identity-admitted family view is UNSAFE on exactly the 181,119 and 324,407 anchors whose excluded carrier has a strict unique maximum, i.e., wherever the direct-carrier view is UNSAFE. The other 9,216 DF anchors with that property are color defects. On the facet task the same equality holds through Π, and the 382 DF and 34,749 FP anchors that only the direct-carrier view certifies are drifted anchors.

Cost per view (Section VI-C). Fig. S3 separates the Path B cost of Fig. 3b by view. At full size the family-3+type view carries 412 of the 669 single-label fibers and 382 ms of the 648 ms of integer-program time. Summed over the Path B cells, single-label fibers grow from 490 to 669 and all fibers from 1,250 to 2,003 while the rows grow tenfold, so new anchors mostly fall into fibers already observed.

External benchmark (Section VI-C). MaDI-Bench is an independent public benchmark. Besides the target schema and taxonomies (Section V), it provides a gold mapping of source columns to target attributes. We audit all five base-tier domains, all six taxonomy-bound attributes and all gold-mapped source columns; products and papers declare no taxonomy transform and are out of scope. Games qualifies only through genres, and companies through both country and industry. The same two domains therefore qualify for every rate threshold up to the genres rate (2,732/74,735) and every count threshold up to the industry count (678). The country query, whose rate (158/13,437) is closest to the 0.01 threshold, does not decide which domains qualify.

An independent implementation, sharing no normalization, classification or aggregation code, recomputes the counts from the source tables and agrees in all 28 checks; every source cell that matches a registered determinant value maps to a single protected value. Since the audit reads source values only, a comparison with the benchmark's normalization gold, which pairs raw source values with expected target values, applies it unchanged. On the audited attributes the gold holds 546 cells; the audit classifies 74 of them as L0 and 39 as L1 and leaves the rest UNVERIFIED, and every classified cell is consistent with its gold value. The derived value equals the complete gold value for 111 classified cells; the other two are genre cells whose gold value lists further genres besides the derived one. Every UNVERIFIED cell is one the gold marks as normalized rather than identical, with a raw value that no registered determinant lists, which is consistent with the L1 counts being lower bounds.

Link-prediction benchmarks (Section VI-C). We audit the published splits of WN18 [1] and WN18RR [2]. For a held-out triple (h, r, t), the audit asks whether the training triples and the registry entail it. A split withholds each held-out

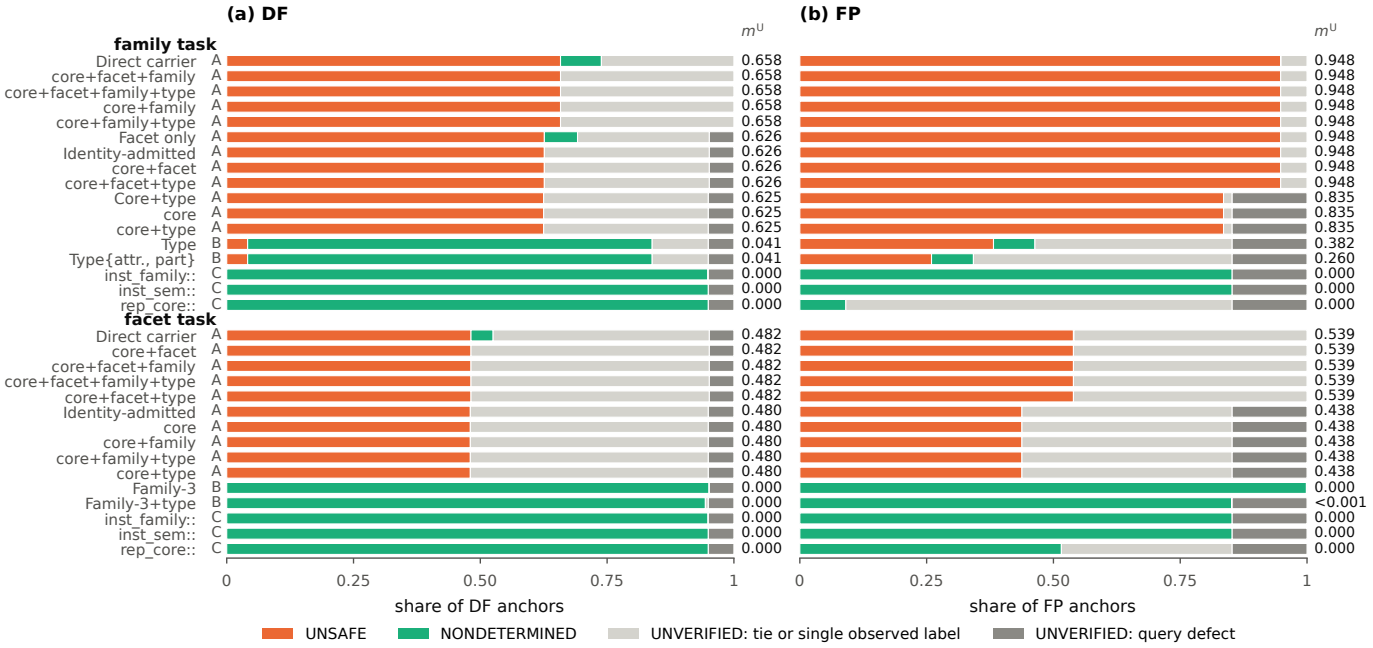


Fig. S1. Verdict composition of all 32 audited views in (a) DF and (b) FP. Rows are ordered by task, then path (letter at the left), then basis (carrier, facet, core). Capitalized names are the direct-carrier, identity-admitted and named views. The number at the right of each bar is m^{U}. Dark gray: anchors outside $O_i^{V,q}$ (query defects); light gray: UNVERIFIED anchors of $O_i^{V,q}$ (a tie or a single observed label).

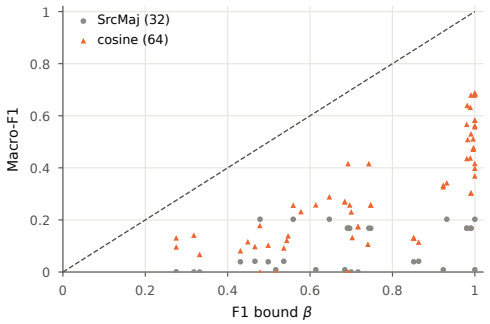


Fig. S2. All 96 probe scores of the audit against their F1 bounds β (Lemma 2), over views, directions and protocols; the dashed line is Macro-F1 $= \beta$.

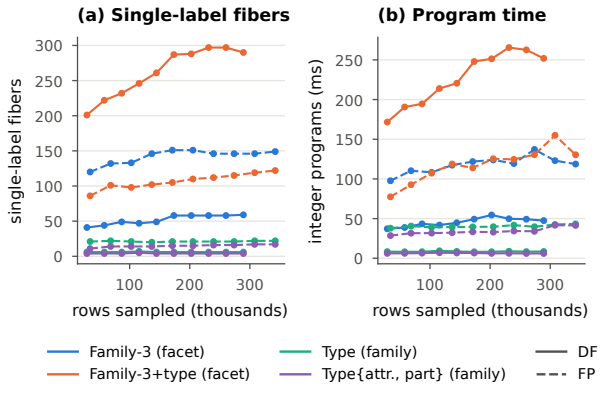


Fig. S3. Path B work per view and dataset on nested hash samples: (a) single-label fibers, each with at most one integer program per rival; (b) integer-program time (median of three runs).

triple by its identity, an analogue of identity-based admission for splits, and admits the rest of the graph to training. The registry is the table of reflexive pointers in the WordNet documentation [3]: a pointer of type P from t to h declares a pointer of type $\bar{P}$ from h to t. For the relations of WN18, the rows cover hypernym and hyponym, instance hypernym and instance hyponym, holonym and meronym, domain of synset to member of domain and the self-reflexive similar-to, verb-group and derivationally-related pointers; also-see has no row and stays outside the registry. A held-out triple (h, r, t) is L1 if training contains a triple (t, P, h) with $\bar{P} = r$, and UNVERIFIED otherwise; a missing partner does not show that a triple is undetermined, so this audit issues no NONDETERMINED verdict. The audit reads no model score, and an independent check confirms all counts (37 checks).

No held-out triple of either release occurs in training. A training triple determines 4,504 of the 5,000 WN18 test triples and 4,512 of its 5,000 validation triples, and 1,052 of the 3,134 WN18RR test triples and 1,046 of its 3,034 validation triples.

The construction script released with WN18RR removes one relation of each of the seven inverse pairs and keeps every other relation, and applied to WN18 it reproduces all three WN18RR splits triple for triple. The L1 test triples of WN18RR therefore all come through self-reflexive pointers (1,011 derivationally related, 38 verb group and three similar to). Our WN18RR count equals the published one [4]; reading the domain row, stated in one direction only, in both directions would add 154 WN18 test and 159 validation triples and give the published WN18 count of 4,658.

WN18 identifiers omit the part of speech. For each of the

11,114 L1 triples of the four splits, the triple and its training witness are WordNet 3.0 pointers in opposite directions between the same two synsets, each with one part of speech; 11,033 match the parts of speech of the WN18 definitions file, and each of the other 81 involves an offset used in multiple part-of-speech files. Of all 151,442 WN18 triples, 151,436 match a pointer by offset and symbol.

D. Released Material

Anchors (Section V). The DF anchors cover 289,202 of the 289,222 images of the Category and Attribute Prediction Benchmark; the other 20 form 10 pairs of paths that differ only in letter case.

The source-release record identifies every input file by SHA-256. The registry step reads a fixed column list without label columns, and the conformance step's access log shows that the label columns are first loaded after FULL-CONFORM has written its output. A verifier that imports no pipeline code recomputes both conformance layers and the fibers of all 32 views from the input view, checks the other main results against the step outputs (89 checks, 83 of them exact, all passing) and confirms the verdict rules on each anchor: every query defect is UNVERIFIED, every query-conforming anchor carries its fiber's verdict, NONDETERMINED holds exactly on the fibers with an observed witness pair, no UNSAFE fiber holds such a pair and Path C issues no UNSAFE verdict. A second check of the support step reads public outputs only (38 certificates, 3,387 basis-block values enumerated).

Two full runs in the pinned environment (Python 3.11.15, NumPy 2.4.4, pandas 3.0.2, SciPy 1.17.1) and with Python 3.11.17 write byte-identical non-timing outputs, all 110 files, the per-anchor ones included.

Build record. The build report (SHA-256 `d569248a...`, in the source-release record) records `conforms: true`, a flag that compares only the number of written rows with the number of matched anchors and that we do not use as evidence. It also records 117,540 dropped core-term entries, which the released rows corroborate: the 66,440 entries of the core block carry the 117,475 facet events of excess that Table V recomputes from the released rows, and the 51,100 representative-core entries equal in number the anchors with an empty representative-core block (Table V). FULL-CONFORM fails on exactly the 65,076 anchors that lost a core entry. On each of them the copied type block equals the type counts of the cleaned core block plus the facet excess counted as attribute events, so T fails on every drifted anchor. The family rule reads the facet and the type: it counted the lost category events as TAX, as W does, and the lost color events as ATTR, whereas W, built from the cleaned type of color, sends color to SURF. Hence W fails exactly on the 13,998 DF anchors with color counts.

E. Support Sensitivity of Path B (Remark 1)

Path B decides each fiber over facet states supported on the released support $S_i = \Pi(K_i)$, the facets that the active core terms of dataset i reach (Definition 2(c)). We compute the verdicts of all 32 views on the released supports and on three counterfactual coordinate sets, with the registry maps, FULL-CONFORM, the observed evidence, the labels, the view projections, the certificate paths and the solver fixed; QUERY-CONFORM is recomputed on each set. The released supports have 16 (DF) and 24 (FP) facets. The counterfactual sets are their union, the 25 facets that active core terms reach; their intersection, with 15 facets; and each released support with color added. On the released supports the computation reproduces every verdict of the paper and every certificate and open fiber of Section A.2. Table S3 lists the Path B results; an independent check verifies each certificate on each set in integer arithmetic, enumerating the facets that the view value bounds and checking that every other supported facet feeds the certified label. Each coordinate set defines its own certification domain, so a verdict that changes between sets answers a different question.

Union. The union adds nine facets to S_{DF} and one, nickname, to S_{FP}. No anchor of any view changes its QUERY-CONFORM flag, and one anchor changes its verdict. The FP family-3+type certificate of value $(0, 0, 1, 2, 1, 0)$ (Table S1) fails, since nickname is an attribute facet that W sends to TAG, so the two attribute events can split between waistline and nickname, and waistline then ties with the category facet.

Intersection. A narrower set changes which anchors are defects, which fibers are certified and one admission decision. Up to 112,630 DF and 263,130 FP anchors that conform on the released support become support defects, and every UNSAFE anchor that loses its verdict is among them, as the support-relative definition of Remark 1 implies. In DF, 115 anchors in 11 family-3+type fibers receive UNSAFE although their fibers on the released support carry a witness pair: the anchors with the other labels become defects, since their evidence uses a facet outside the intersection. The UNVERIFIED mass of family-3 rises to 0.438 (DF) and 0.770 (FP), so this view, admitted on the released support under a 0.05 policy, is rejected.

Color. No active core term reaches color, although 13,998 DF anchors materialize it. The family-task Path B views expose only `type::` counts, which omit color, and W sends color to SURF (Table S2). Once color is added, every fiber of these views holds states with arbitrarily many color events and an unchanged view value, at which SURF exceeds the certified label (Theorem 1(a)). The computation confirms this: all nine family-task certificates fail and their 243,357 view–anchor pairs become UNVERIFIED, while the family-3+type certificate, whose view fixes SURF, remains. The only QUERY-CONFORM changes are in the five facet-task views that expose the carrier and need no registered map: there the 13,998 DF anchors that materialize color pass QUERY-CONFORM, and these views stay rejected.

Admission. On the union and with color, all admission decisions of all views stay the same for each ϵ below 0.148: a voided certificate leaves its mass UNVERIFIED, and the family-task views keep UNVERIFIED mass of at least 0.202. For larger ϵ, the family-3+type view would be admitted on the union,

TABLE S3

PATH B ON THE RELEASED SUPPORTS AND ON THE THREE COUNTERFACTUAL COORDINATE SETS OF SECTION E: CERTIFICATES (CERT.), UNSAFE MASS m^{U} AND UNVERIFIED MASS m^{V} PER VIEW AND DATASET. LAST ROW: ALL PATH B CERTIFICATES / SINGLE-LABEL FIBERS THAT STAY OPEN, EACH CHECKED EXACTLY. THE REGISTRY MAPS, FULL-CONFORM, THE EVIDENCE, THE LABELS, THE VIEW PROJECTIONS, THE PATHS AND THE SOLVER ARE FIXED; QUERY-CONFORM IS RECOMPUTED ON EACH SET.

View	Set	Released			Union			Intersection			Released + color		
		Cert.	m^{U}	m^{V}	Cert.	m^{U}	m^{V}	Cert.	m^{U}	m^{V}	Cert.	m^{U}	m^{V}
Type (family)	DF	1	0.0412	0.1612	1	0.0412	0.1612	1	0.0412	0.5312	0	0.0000	0.2024
Type (family)	FP	6	0.3818	0.5365	6	0.3818	0.5365	3	0.0009	0.9175	0	0.0000	0.9183
Type{attr., part} (family)	DF	1	0.0412	0.1611	1	0.0412	0.1611	1	0.0412	0.5308	0	0.0000	0.2024
Type{attr., part} (family)	FP	1	0.2596	0.6577	1	0.2596	0.6577	1	0.0004	0.9170	0	0.0000	0.9174
Family-3 (facet)	DF	0	0.0000	0.0489	0	0.0000	0.0489	0	0.0000	0.4384	0	0.0000	0.0489
Family-3 (facet)	FP	0	0.0000	0.0012	0	0.0000	0.0012	0	0.0000	0.7701	0	0.0000	0.0012
Family-3+type (facet)	DF	0	0.0000	0.0562	0	0.0000	0.0562	11	0.0004	0.4398	0	0.0000	0.0562
Family-3+type (facet)	FP	1	$2.9{\cdot}10^{-6}$	0.1481	0	0.0000	0.1481	1	$2.9{\cdot}10^{-6}$	0.9170	1	$2.9{\cdot}10^{-6}$	0.1481
All Path B views			10 / 659			9 / 660			18 / 411			1 / 668	

where it has no UNSAFE anchor, and with color the two type views, whose certificates color voids, would be admitted once ϵ reaches their UNVERIFIED mass.

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